

React HandsOn:

Document 1: ReactJS-HOL 1

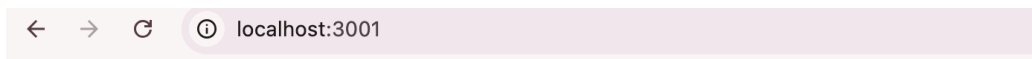
← → ↻ ⓘ localhost:3000

Welcome to the first session of React

```
✓ REACT-HOL
  ✓ myfirstreact
    > node_modules
    > public
    ✓ src
      # App.css
      JS App.js
      JS App.test.js
      # index.css
      JS index.js
      logo.svg
      JS reportWebVitals.js
      JS setupTests.js
      .gitignore
      {} package-lock.json
      {} package.json
      ⓘ README.md
```

```
myfirstreact > src > JS App.js > ...
1  import logo from './logo.svg';
2  import './App.css';
3
Windsurf: Refactor | Explain | Generate JSDoc | X
4  function App() {
5      return (
6          <h1>Welcome to the first session of React</h1>
7      );
8  }
9
10 export default App;
11
```

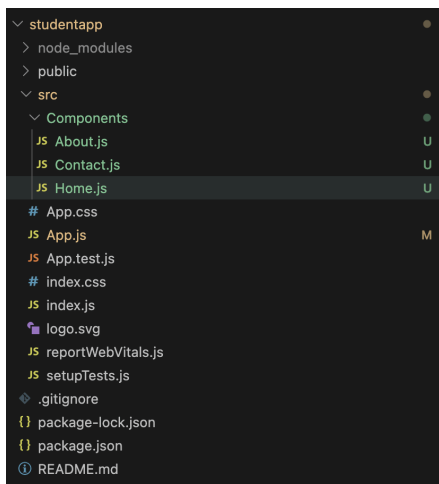
Document 2:



Welcome to the Home page of Student Management Portal

Welcome to the About page of Student Management Portal

Welcome to the Contact page of Student Management Portal



```

1  import './App.css';
2
3  import Home from './Components/Home';
4  import About from './Components/About';
5  import Contact from './Components/Contact';
6
7  Windsurf: Refactor | Explain | Generate JSDoc | X
8  function App() {
9
10     return (
11         <div className="container">
12             <Home />
13             <About />
14             <Contact />
15         </div>
16     );
17
18     };
19
20     export default App;

```

```

studentapp > src > Components > JS About.js > [⌕] default
1  import React, { Component } from "react";
2
3  Windsurf: Refactor | Explain
4
5  Windsurf: Refactor | Explain | Generate JSDoc | X
6  class About extends Component {
7
8      render() {
9          return (
10             <div>
11                 <h2>Welcome to the About page of Student Management Po
12             </div>
13         );
14     }
15
16     export default About;

```

```
udentapp > src > Components > JS Contact.js > JS Contact
1  import React, { Component } from "react";
2
Windsurf: Refactor | Explain
3  class Contact extends Component {
4
Windsurf: Refactor | Explain | Generate JSDoc | X
5      render() {
6          return (
7              <div>
8                  <h2>Welcome to the Contact page of Student Management
9              </div>
10          );
11      }
12
13  }
14
15  export default Contact;
```

```
1  import React, { Component } from "react";
2
Windsurf: Refactor | Explain
3  class Home extends Component {
4
Windsurf: Refactor | Explain | Generate JSDoc | X
5      render() {
6          return (
7              <div>
8                  <h2>Welcome to the Home page of Student Management Portal</h2>
9              </div>
10          );
11      }
12
13  }
14
15  export default Home;
```

ReactJS-HOL 3: Score Calculator App

localhost:3001

Student Score Card

Name : Sindhu

School : ABC Public School

Total Marks : 450

Goal : 5

Average Score : 90

```
import "../Stylesheets/mystyle.css";

Windsurf: Refactor | Explain | Generate JSDoc | X
function CalculateScore(props) {

  const average = props.total / props.goal;

  return (
    <div className="container">
      <div className="card">
        <h2>Student Score Card</h2>
        <h3>Name : {props.name}</h3>
        <h3>School : {props.school}</h3>
        <h3>Total Marks : {props.total}</h3>
        <h3>Goal : {props.goal}</h3>
        <h3>Average Score : {average}</h3>
      </div>
    </div>
  );
}

export default CalculateScore;
```

```

.container{
  text-align:center;
  margin-top:40px;
}

.card{
  border:2px solid gray;
  width:350px;
  margin:auto;
  padding:20px;
  border-radius:10px;
  background-color:#f4f4f4;
}

h2{
  color:blue;
}

```

```

src > src > App.js > {} default
import CalculateScore from "../Components/CalculateScore";

Windsurf: Refactor | Explain | Generate JSDoc | X
function App() {

  return (

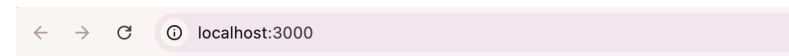
    <CalculateScore
      name="Sindhu"
      school="ABC Public School"
      total={450}
      goal={5}
    />

  );
}

export default App;

```

EXERCISE 3:



Blog Posts

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qui est esse

est rerum tempore vitae sequi sint nihil reprehenderit dolor beatae ea dolores neque fugiat bla

ea molestias quasi exercitationem repellat qui ipsa sit aut

et iusto sed quo iure voluptatem occaecati omnis eligendi aut ad voluptatem doloribus vel acc

eum et est occaecati

ullam et saepe reiciendis voluptatem adipisci sit amet autem assumenda provident rerum culp

nesciunt quas odio

repudiandae veniam quaerat sunt sed alias aut fugiat sit autem sed est voluptatem omnis pos

dolorem eum magni eos aperiam quia

```
> JS App.js > [🔍] default
import Posts from "./Posts";

Windsurf: Refactor | Explain | Generate JSDoc | ✕
function App() {
  return (
    <div>
      <Posts />
    </div>
  );
}

export default App;
```

src > JS Post.js > Post

Windsurf: Refactor | Explain

```
1 class Post {
2
3     Windsurf: Refactor | Explain | Generate JSDoc | X
4     constructor(id, title, body) {
5         this.id = id;
6         this.title = title;
7         this.body = body;
8     }
9 }
10
11 export default Post;
```

src > JS Posts.js > Posts

```
import React, { Component } from "react";
import Post from "../Post";
```

Windsurf: Refactor | Explain

```
class Posts extends Component {
```

Windsurf: Refactor | Explain | Generate JSDoc | X

```
    constructor() {
```

```
        super();
```

```
        this.state = {
            posts: []
        };
    }
```

Windsurf: Refactor | Explain | Generate JSDoc | X

```
    loadPosts() {
```

```
        fetch("https://jsonplaceholder.typicode.com/posts")
            .then(response => response.json())
            .then(data => {
                const posts = data.map(
                    item => new Post(item.id, item.title, item.body)
                );
                this.setState({
                    posts: posts
                });
            });
    }
```



```
class Posts extends Component {  
  }
```

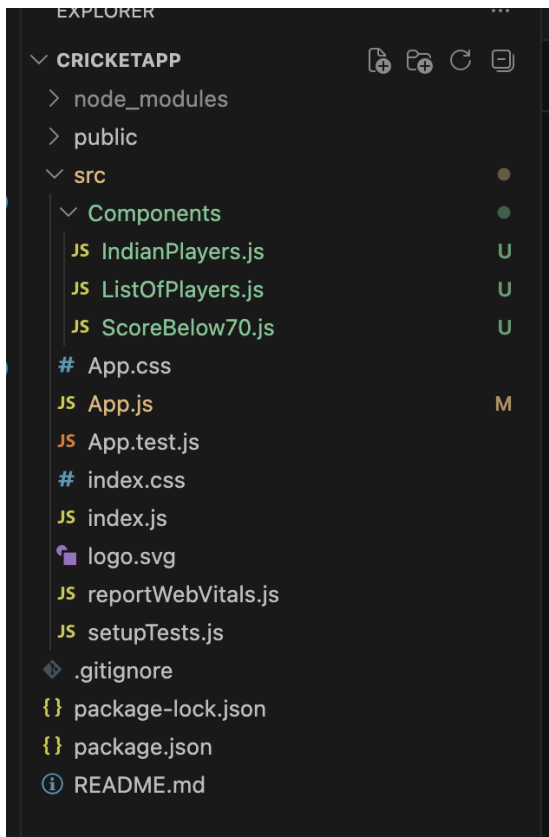
Windsurf: Refactor | Explain | Generate JSDoc | X

```
  componentDidCatch(error) {  
    alert(error);  
  }
```

Windsurf: Refactor | Explain | Generate JSDoc | X

```
  render() {  
  
    return (  
      <div>  
        <h1>Blog Posts</h1>  
        {  
          this.state.posts.map(post => (  
            <div key={post.id}>  
              <h3>{post.title}</h3>  
              <p>{post.body}</p>  
              <hr />  
            </div>  
          ))  
        }  
      </div>  
    );  
  }
```

ReactJS-HOL 5:



Windsurf: Refactor | Explain | Generate JSDoc | X

```
export function OddPlayers(players) {  
  
  const [first, , third, , fifth] = players;  
  
  return (  
    <div>  
      <p>First : {first}</p>  
  
      <p>Third : {third}</p>  
  
      <p>Fifth : {fifth}</p>  
    </div>  
  );  
}
```

Windsurf: Refactor | Explain | Generate JSDoc | X

```
export function EvenPlayers(players) {  
  
  const [, second, , fourth, , sixth] = players;  
  
  return (  
    <div>  
      <p>Second : {second}</p>  
  
      <p>Fourth : {fourth}</p>  
  
      <p>Sixth : {sixth}</p>  
    </div>  
  );  
}
```

Windsurf: Refactor | Explain | Generate JSDoc | X

```
export function ListOfIndianPlayers({ players }) {  
  
  return (  
    <ul>  
      {  
        players.map((player, index) => (  
          <li key={index}>{player}</li>  
        ))  
      }  
    </ul>  
  );  
}
```

```
src > Components > JS ListOfPlayers.js > [🔍] default
Windsurf: Refactor | Explain | Generate JSDoc | X
1 function ListOfPlayers({ players }) {
2
3     return (
4
5         <div>
6
7             {
8
9                 players.map(player => (
10
11                     <li key={player.id}>
12
13                         Mr. {player.name} - {player.score}
14
15                     </li>
16
17                 ))
18
19             }
20
21         </div>
22
23     );
24
25 }
26
27 export default ListOfPlayers;
```

```
Windsurf: Refactor | Explain | Generate JSDoc | X
1 function ScoreBelow70({ players }) {
2
3     const filteredPlayers = players.filter(player => player.score < 70);
4
5     return (
6
7         <div>
8
9             {
10
11                 filteredPlayers.map(player => (
12
13                     <li key={player.id}>
14
15                         Mr. {player.name} - {player.score}
16
17                     </li>
18
19                 ))
20
21             }
22
23         </div>
24
25     );
26
27 }
28
29 export default ScoreBelow70;
```

```

JS App.js > App
import ListOfPlayers from "../Components/ListOfPlayers";
import ScoreBelow70 from "../Components/ScoreBelow70";
import {
  OddPlayers,
  EvenPlayers,
  ListOfIndianPlayers
} from "../Components/IndianPlayers";

Windsurf: Refactor | Explain | Generate JSDoc | X
function App() {

  const flag = true;

  const players = [
    { id: 1, name: "Jack", score: 50 },
    { id: 2, name: "Michael", score: 70 },
    { id: 3, name: "John", score: 40 },
    { id: 4, name: "Ann", score: 61 },
    { id: 5, name: "Elisabeth", score: 61 },
    { id: 6, name: "Sachin", score: 95 },
    { id: 7, name: "Dhoni", score: 100 },
    { id: 8, name: "Virat", score: 84 },
    { id: 9, name: "Jadeja", score: 64 },
    { id: 10, name: "Raina", score: 75 },
    { id: 11, name: "Rohit", score: 80 }
  ];

  const indianTeam = [
    "Sachin",
    "Dhoni",
    "Virat",
    "Rohit",
    "Yuvraj",
    "Raina"
  ];

  const t20Players = [
    "First Player",
    "Second Player",
    "Third Player"
  ];

```

```

9 function App() {
60   <hr />
61
62   <h1>Players having Score less than 70</h1>
63
64   <ScoreBelow70 players={players} />
65
66   </div>
67
68   );
69
70 }
71
72 return (
73
74   <div>
75     <h1>Odd Players</h1>
76
77     {OddPlayers(indianTeam)}
78
79     <hr />
80
81     <h1>Even Players</h1>
82
83     {EvenPlayers(indianTeam)}
84
85     <hr />
86
87     <h1>Merged Indian Players</h1>
88
89     <ListOfIndianPlayers players={mergedPlayers} />
90
91   </div>
92
93   );
94
95
96 }

```

ReactJS-HOL 10: Office Space Rental App (JSX):

← → ↻ ⓘ localhost:3001

Office Space, at Affordable Range



Name: DBS

Rent: Rs. 50000

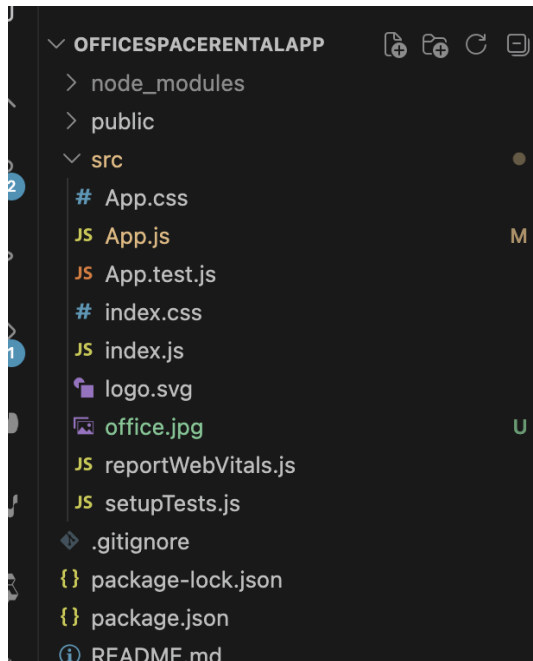
Address: Chennai



Name: Infosys

Rent: Rs. 75000

Address: Bangalore



```
import office from "./office.jpg";

function App() {

  const offices = [

    {
      id: 1,
      name: "DBS",
      rent: 50000,
      address: "Chennai"
    },

    {
      id: 2,
      name: "Infosys",
      rent: 75000,
      address: "Bangalore"
    },

    {
      id: 3,
      name: "TCS",
      rent: 55000,
      address: "Hyderabad"
    }
  ];

  return (

    <div style={{ margin: "40px" }}>

      <h1>Office Space, at Affordable Range</h1>

      {
        offices.map((officeItem) => (

          <div key={officeItem.id} style={{ marginBottom: "40px" }}>

            <img
              src={office}
            />
          </div>
        ))
      }
    </div>
  );
}
```

```

3   function App() {
36   }
37
38   offices.map((officeItem) => (
39
40     <div key={officeItem.id} style={{ marginBottom: "40px" }}>
41
42       <img
43         src={office}
44         alt="Office"
45         width="250"
46       />
47
48       <h2>Name: {officeItem.name}</h2>
49
50       <h3
51         style={{
52           color:
53             officeItem.rent <= 60000
54             ? "red"
55             : "green"
56         }}
57       >
58         Rent: Rs. {officeItem.rent}
59       </h3>
60
61       <h3>
62         Address: {officeItem.address}
63       </h3>
64     </div>
65   ))
66
67   }
68
69 }
70
71 </div>
72
73 );
74
75 }
76
77 export default App;

```


HOL 11:

← → ↻ localhost:3001

0

Increment

Decrement

Say Welcome

Click on me

Currency Converter!!!

Amount

Currency

Submit

Currency Converter!!!

Amount

Currency

Submit

```

src > JS App.js > App > offices > rent
1 import office from "./office.jpg";
2
3 function App() {
4
5     const offices = [
6
7         {
8             id: 1,
9             name: "DBS",
10            rent: 50000,
11            address: "Chennai"
12        },
13
14        {
15            id: 2,
16            name: "Infosys",
17            rent: 75000,
18            address: "Bangalore"
19        },
20
21        {
22            id: 3,
23            name: "TCS",
24            rent: 55000,
25            address: "Hyderabad"
26        }
27    ];
28
29    return (
30
31        <div style={{ margin: "40px" }}>
32
33            <h1>Office Space, at Affordable Range</h1>
34
35            {
36
37                offices.map((officeItem) => (

```

```

3 function App() {
38     offices.map((officeItem) => (
39
40         <div key={officeItem.id} style={{ marginBottom: "40px" }}>
41
42             <img
43                 src={office}
44                 alt="Office"
45                 width="250"
46             />
47
48             <h2>Name: {officeItem.name}</h2>
49
50             <h3
51                 style={{
52                     color:
53                         officeItem.rent <= 60000
54                         ? "red"
55                         : "green"
56                 }}
57             >
58                 Rent: Rs. {officeItem.rent}
59             </h3>
60
61             <h3>
62                 Address: {officeItem.address}
63             </h3>
64
65             </div>
66
67         ))
68
69     }
70
71 </div>
72
73 );
74
75 }
76
77 export default App;

```

HOL 12:

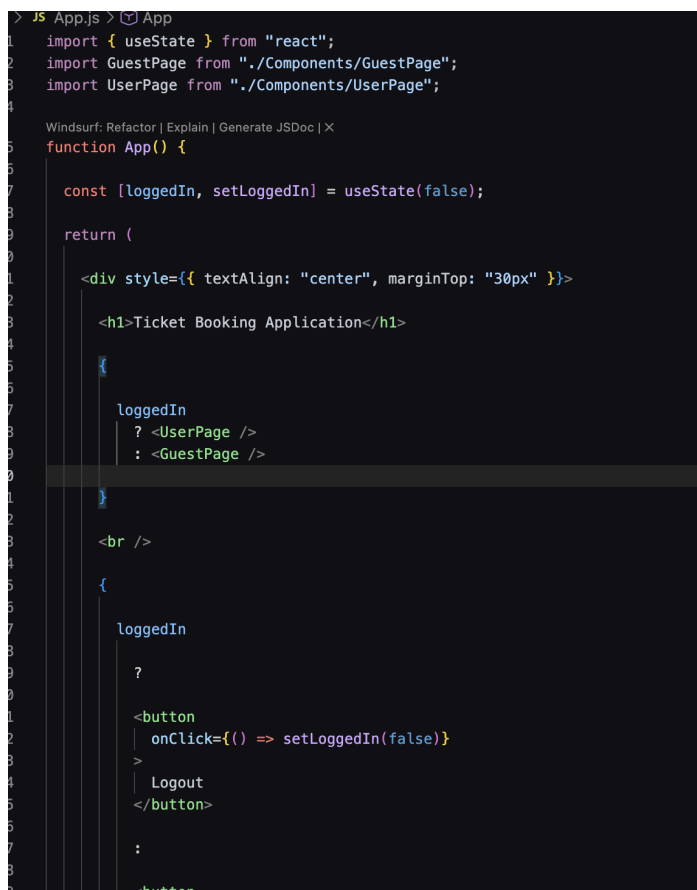
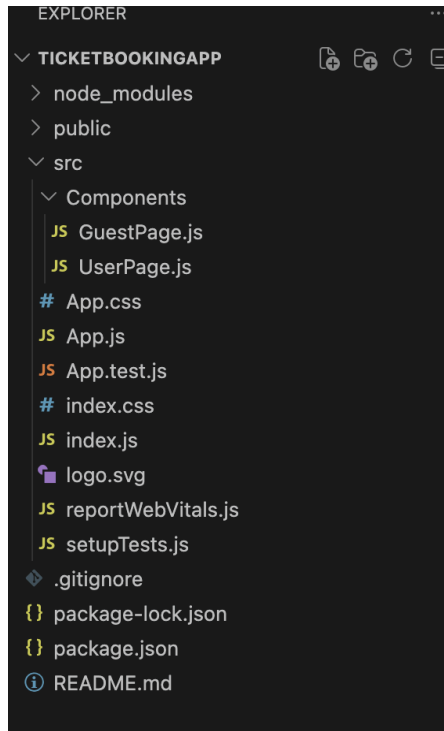
Ticket Booking Application

Flight Details

Please Login to Book Tickets

Login

Flight	From	To	Price
AI101	Hyderabad	Bangalore	₹4500
AI202	Chennai	Mumbai	₹5500



```
function App() {  
  : <GuestPage />  
}  
  
<br />  
  
{  
  loggedIn  
  ?  
  <button  
    | onClick={() => setLoggedIn(false)}  
  >  
    | Logout  
  </button>  
  :  
  <button  
    | onClick={() => setLoggedIn(true)}  
  >  
    | Login  
  </button>  
}  
</div>  
);  
}  
  
export default App;
```

ReactJS-HOL 13: Blogger App (Conditional Rendering & Lists)

← → ↻ ⓘ localhost:3003

Blogger Application

Book Details

- Java Programming - ₹650
 - React Essentials - ₹750
 - Spring Boot - ₹820
-

Blog Details

- React Basics
 - Understanding JSX
 - React Hooks
-

Course Details

- Java
- Spring Boot
- React
- SQL

```

import BookDetails from "../Components/BookDetails";
import BlogDetails from "../Components/BlogDetails";
import CourseDetails from "../Components/CourseDetails";

```

Windsurf: Refactor | Explain | Generate JSDoc | X

```

function App() {

  const showBooks = true;

  const showBlogs = true;

  const showCourses = true;

  return (

    <div style={{ margin: "30px" }}>

      <h1>Blogger Application</h1>

      {/* Conditional Rendering using && */}

      {showBooks && <BookDetails />}

      <hr />

      {/* Conditional Rendering using Ternary */}

      {

        showBlogs

        ? <BlogDetails />

        : <h2>No Blogs Available</h2>

      }

      <hr />

      {/* Conditional Rendering using Element Variable */}

```

```

App.js > App
function App() {

  : <h2>No Blogs Available</h2>

}

<hr />

{/* Conditional Rendering using Element Variable */}

{

  (() => {

    let component;

    if (showCourses) {

      component = <CourseDetails />;

    } else {

      component = <h2>No Courses Available</h2>;

    }

    return component;

  })()

}

</div>

);

}

export default App;

```

```

function BlogDetails() {
  const blogs = [
  ];

  return (
    <div>
      <h2>Blog Details</h2>

      <ul>
        {
          blogs.map(blog => (
            <li key={blog.id}>
              {blog.title}
            </li>
          ))
        }
      </ul>
    </div>
  );
}

export default BlogDetails;

```

Windows: Reactor | Explain | Generate JSBoc | X

```

function BookDetails() {

  const books = [

    {
      id: 1,
      name: "Java Programming",
      price: 650
    },

    {
      id: 2,
      name: "React Essentials",
      price: 750
    },

    {
      id: 3,
      name: "Spring Boot",
      price: 820
    }
  ];

  return (
    <div>

```



```

    return (
      <div>
        <h2>Book Details</h2>
        <ul>
          {
            books.map(book => (
              <li key={book.id}>
                {book.name} - ₹{book.price}
              </li>
            ))
          }
        </ul>
      </div>
    );
  }
}

export default BookDetails;

```

Windsurf: Refactor | Explain | Generate JSDoc | X

```

function CourseDetails() {
  const courses = [
    "Java",
    "Spring Boot",
    "React",
    "SQL"
  ];

  return (
    <div>
      <h2>Course Details</h2>
      <ul>
        {
          courses.map((course, index) => (
            <li key={index}>
              {course}
            </li>
          ))
        }
      </ul>
    </div>
  );
}

```

```

function CourseDetails() {
  const courses = [
  ];

  return (
    <div>
      <h2>Course Details</h2>
      <ul>
        {
          courses.map((course, index) => (
            <li key={index}>
              {course}
            </li>
          ))
        }
      </ul>
    </div>
  );
}

export default CourseDetails;

```

```

> JS App.js > App
1  import BookDetails from "../Components/BookDetails";
2  import BlogDetails from "../Components/BlogDetails";
3  import CourseDetails from "../Components/CourseDetails";
4
Windsurf: Refactor | Explain | Generate JSDoc | X
5  function App() {
6
7    const showBooks = true;
8
9    const showBlogs = true;
10
11    const showCourses = true;
12
13    return (
14      <div style={{ margin: "30px" }}>
15        <h1>Blogger Application</h1>
16
17        /* Conditional Rendering using && */
18
19        {showBooks && <BookDetails />}
20
21        <hr />
22
23        /* Conditional Rendering using Ternary */
24
25        {
26          showBlogs
27            ? <BlogDetails />
28            : <h2>No Blogs Available</h2>
29        }
30
31        <hr />
32
33        /* Conditional Rendering using Element Variable */

```

```
function App() {  
  <nr />  
  
  { /* Conditional Rendering using Element Variable */  
  {  
    (() => {  
      let component;  
  
      if (showCourses) {  
        component = <CourseDetails />;  
      } else {  
        component = <h2>No Courses Available</h2>;  
      }  
  
      return component;  
    })()  
  }  
  </div>  
  );  
}  
  
export default App;
```